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|---|---------|--|----|-----|
| 1 | (a) (i) | micrometer (screw gauge) / travelling microscope | B1 | [1] |
| | (ii) | <i>either</i> ohm-meter or voltmeter and ammeter
or multimeter/avo on ohm setting | B1 | [1] |
| | (iii) | <i>either</i> (calibrated) c.r.o. or a.c. voltmeter and $\times \sqrt{2}$ | B1 | [1] |
| | (b) | density = mass / volume | C1 | |
| | | = $580 / 6^3 = 2.685 \text{ g cm}^{-3}$...(<i>allow 2.68, 2.69, 2.7</i>) | A1 | |
| | | % uncertainty in mass = $(10 / 580) \times 100 = 1.7\%$ | C1 | |
| | | % uncertainty in volume = $3 \times (0.1 / 6) \times 100 = 5.0\%$ | C1 | |
| | | uncertainty in density = 0.18 g cm^{-3} | | |
| | | density = $2.7 \pm 0.2 \text{ g cm}^{-3}$ | A1 | [5] |
| | | (<i>answer $2.69 \pm 0.09 \text{ g cm}^{-3}$ scores 4 marks</i>) | | |
| 2 | (a) | ball moving in opposite direction (after collision) | B1 | [1] |